PREDICTIVE MONITORING OF AI-GENERATED HEALTHCARE CONTENT SATURATION AND MISINFORMATION RISK

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**PREDICTIVE MONITORING OF AI-GENERATED HEALTHCARE CONTENT SATURATION AND MISINFORMATION RISK**" has been carried out by the students **SHRUTHIKA K, SRIDARSENI K P, SWETHA M**, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

The rapid growth of digital platforms and artificial intelligence has led to an increase in online healthcare information, making it difficult to verify its reliability. This project proposes a system titled Predictive Monitoring of AI-Generated Healthcare Content Saturation and Misinformation Risk to analyze and evaluate the credibility of medical articles. The system uses web scraping techniques to collect article content from user-provided URLs and processes it using Natural Language Processing and machine learning methods. It performs text preprocessing, medical claim detection, and AI-generated content identification to detect misleading or suspicious information. The system identifies important medical entities and flags exaggerated or unsupported health claims. It also estimates the probability of AI-generated content based on linguistic patterns. The results are presented in the form of credibility indicators and summarized insights to assist users. This approach helps users make informed decisions and reduces the spread of healthcare misinformation in digital platforms.

Keywords: Healthcare Misinformation, AI-Generated Content, Natural Language Processing, Machine Learning, Web Scraping, Medical Claim Detection, Credibility Analysis, Text Analysis.

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SIGNATURE

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TABLE OF CONTENTS

CHAPTER	TITLE	PAGE No.
	ABSTRACT	**iii**
	LIST OF FIGURES	**ix**
	LIST OF ABBREVIATIONS	**x**
1	**INTRODUCTION**	**1**
	1.1 DESCRIPTION	1
	1.2 HEALTH CONTENT ANALYSIS	2
	1.3 DATA ANALYTICS FOR MISINFORMATION RISK ASSESSMENT	2
2	**LITERATURE SURVEY**	**4**
3	**EXISTING SYSTEM**	**15**
	3.1 DISADVANTAGES	17
4	**PROBLEMS IDENTIFIED**	**18**
5	**PROPOSED SYSTEM**	**19**
	5.1 ADVANTAGES	21
6	**SYSTEM REQUIREMENTS**	**22**
	6.1 HARDWARE REQUIREMENTS	22
	6.2 SOFTWARE REQUIREMENTS	22
7	**SYSTEM IMPLEMENTATIONS**	**23**
	7.1 LIST OF MODULES	23
	7.2 MODULES DESCRIPTION	23
	7.2.1 User Interface Module	23
	7.2.2 Data Collection Module	24
	7.2.3 Text Preprocessing Module	24
	7.2.4 Medical Claim Detection Module	24

	7.2.5 AI Content Detection Module	25
	7.2.6 Result Presentation Module	25
8	**SYSTEM TESTING**	**26**
	8.1 UNIT TESTING	26
	8.2 INTEGRATION TESTING	26
	8.3 FUNCTIONAL TESTING	27
	8.4 PERFORMANCE TESTING	28
	8.5 USABILITY TESTING	28
9	**RESULTS AND DISCUSSION**	**30**
10	**CONCLUSION AND FUTURE ENHANCEMENTS**	**34**
	10.1 CONCLUSION	34
	10.2 FUTURE ENHANCEMENTS	35
	APPENDIX A - SOURCE CODE	**39**
	APPENDIX B - SCREENSHOTS	**47**
	REFERENCES	**50**

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	Existing System Diagram	16
5.1	Proposed System Diagram	20
9.1	Performance comparison	31
9.2	Accuracy comparison	32

LIST OF SYMBOLS AND ABBREVIATIONS

AI	-	Artificial Intelligence
ML	-	Machine Learning
NLP	-	Natural Language Processing
LLM	-	Analysis of Variance
NER	-	Named Entity Recognition
DOI	-	Digital Object Identifier
BERT	-	Bidirectional Encoder Representations from Transformers
API	-	Applicant Programming Interface
HTML	-	Hyper Text Markup Language
CSS	-	Cascading Style Sheets
HTTP	-	Hyper Text Transfer Protocol
JSON	-	JavaScript Object Notation
SQL	-	Structured Query Language
UI	-	User Interface
IDE	-	Integrated Development Environment
EAI	-	Explainable Artificial Intelligence
SHAP	-	Shapley Additive Explanations
LIME	-	Local Interpretable Model Agnostic Explanations

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

In the digital era, healthcare information is widely accessed through online news portals, blogs, research platforms, and social media. While this accessibility improves awareness and knowledge sharing, it also increases the risk of misinformation spreading rapidly across the internet. With the advancement of Artificial Intelligence (AI), large language models are now capable of generating highly realistic healthcare articles, summaries, and medical explanations. Although these AI systems provide efficiency and scalability, they may also produce content that lacks proper verification, scientific validation, or reliable references. This creates challenges in distinguishing authentic medical information from AI-generated or misleading content. Healthcare misinformation can influence public opinion, encourage unsafe treatments, and reduce trust in medical institutions. Traditional manual fact-checking methods are insufficient to monitor the massive volume of digital content published daily. Therefore, there is a strong need for an intelligent, automated system capable of analyzing healthcare articles, detecting AI-generated patterns, evaluating credibility, and predicting misinformation risk. The Predictive Monitoring of AI-Generated Healthcare Content Saturation and Misinformation Risk system was developed to address this challenge. By combining Natural Language Processing (NLP), machine learning, biomedical entity recognition, and predictive risk scoring, the system provides real-time content analysis and credibility assessment. This project aims to enhance digital healthcare safety by ensuring accurate, reliable, and trustworthy medical information for users.

1.2 HEALTHCARE CONTENT ANALYSIS

Artificial Intelligence plays a crucial role in analyzing large volumes of healthcare-related text data efficiently and accurately. In the context of content monitoring, AI techniques such as machine learning, deep learning, and transformer-based models are used to classify, summarize, and evaluate medical articles. These models analyze linguistic patterns, contextual relationships, and structural features to determine whether the content is likely human-written or AI-generated.

AI systems can also identify suspicious claims, exaggerated medical statements, and unsupported treatment recommendations. By training on large datasets, machine learning algorithms learn to recognize patterns associated with misinformation. Transformer models further enhance contextual understanding, enabling better detection of complex healthcare narratives.

Additionally, AI-based summarization tools help extract key insights from long articles and highlight potential inconsistencies. The integration of AI ensures automated, scalable, and real-time monitoring of digital healthcare content. Unlike manual verification methods, AI-driven systems continuously adapt to new writing styles and evolving misinformation techniques. Through predictive analysis, the system assigns credibility scores and evaluates the likelihood of AI-generated saturation. This intelligent approach improves accuracy, reduces human workload, and strengthens healthcare information reliability across digital platforms.

1.3 DATA ANALYTICS FOR MISINFORMATION RISK ASSESSMENT

Data analytics plays an essential role in identifying patterns, trends, and risk factors associated with AI-generated healthcare content. By analyzing large datasets of online medical articles, predictive models can detect content saturation levels and evaluate misinformation probability. Techniques such as statistical analysis, classification algorithms, anomaly detection, and trend forecasting are used to

examine textual features, source reliability, and linguistic indicators. Data-driven systems can measure content duplication, analyze publication frequency, and identify sudden increases in AI-generated material. Visualization tools such as dashboards and risk heatmaps help present credibility scores in an understandable format. Machine learning models trained on labeled datasets can predict whether a healthcare article contains misleading information. Additionally, integration of biomedical entity recognition allows the system to verify diseases, treatments, and medical terms mentioned in the content. Data analytics also supports DOI verification and reference validation to ensure scientific authenticity. By converting raw textual data into meaningful insights, the system provides a structured risk assessment report. This predictive approach enables early detection of misinformation trends, helping users, researchers, and healthcare professionals make informed decisions. Ultimately, data analytics enhances the system's ability to monitor content saturation, reduce misinformation risks, and maintain trust in digital healthcare information.

CHAPTER 2

LITERATURE SURVEY

2.1 HYBRID DEEP LEARNING FRAMEWORK FOR FAKE NEWS DETECTION USING LSTM-CGPNN

R. Ayyasamy and C. Ponnusamy (2025) proposed a hybrid deep learning architecture aimed at improving the detection of fake news across digital platforms. The authors focused on addressing the increasing spread of misinformation through social media and online news websites, which has become a major challenge for digital information ecosystems. Their framework integrates Long Short-Term Memory (LSTM) networks with a Convolutional Gaussian Perceptron Neural Network (CGPNN) to effectively capture both contextual and sequential relationships within textual data.

The LSTM component was used to analyze sequential dependencies between words and sentences in news articles, enabling the model to understand contextual patterns that may indicate misleading or fabricated content. The CGPNN classifier enhanced the feature extraction process by improving the representation of textual data in high-dimensional space. Additionally, the authors applied metaheuristic optimization techniques to tune model parameters and improve classification accuracy.

Experimental evaluation showed that the proposed hybrid model outperformed traditional machine learning classifiers and standalone deep learning models such as CNN and LSTM. The system achieved higher accuracy in identifying misleading content by analyzing linguistic patterns, semantic structures, and contextual inconsistencies in news articles. However, the authors acknowledged that the model required large datasets and significant computational resources for training. They also

noted that detecting domain-specific misinformation, such as healthcare-related false claims, requires specialized datasets and domain knowledge.

2.2 DEEP LEARNING-BASED HEALTHCARE MISINFORMATION DETECTION IN SOCIAL MEDIA

A. Banerjee and S. Gupta (2025) developed a deep learning-based framework designed to identify healthcare-related misinformation circulating on social media platforms. The rapid growth of digital communication has significantly increased the accessibility of health information online. While this accessibility benefits users seeking medical knowledge, it also increases the risk of misinformation spreading quickly across online communities. Misleading health information related to treatments, medications, or disease prevention can lead to serious consequences for individuals and public health systems.

The researchers proposed a Natural Language Processing-based framework that utilizes transformer-based language models such as BERT and BioBERT to analyze textual health discussions on social media platforms. These models were specifically chosen because they are capable of understanding contextual relationships within text and can handle complex medical terminology more effectively than traditional machine learning approaches. The system also integrates sentiment analysis and topic modeling techniques to identify emotional or persuasive language often associated with misinformation campaigns.

To evaluate the effectiveness of their model, the authors collected large datasets from online platforms including discussion forums, blogs, and health-related social media posts. The datasets were carefully labeled to distinguish between verified medical information and misleading health claims. After preprocessing the data through tokenization, stop-word removal, and lemmatization, the deep learning models were trained to classify content based on credibility.

The experimental results showed that transformer-based models significantly improved the accuracy of misinformation detection compared to classical machine learning algorithms such as Logistic Regression and Support Vector Machines. However, the researchers identified several challenges that still limit the effectiveness of automated healthcare misinformation detection systems. These include dataset imbalance, rapidly evolving misinformation strategies, and the difficulty of verifying complex medical claims without professional expertise. The authors concluded that future systems should integrate knowledge graphs and trusted medical databases to improve verification accuracy.

2.3 AI-BASED CREDIBILITY ASSESSMENT OF ONLINE HEALTH INFORMATION

J. Carter and L. Mitchell (2025) proposed an artificial intelligence-based system designed to assess the credibility of health information published on the internet. The authors emphasized that individuals frequently rely on online sources such as blogs, news websites, and social media platforms to obtain health-related advice. Unfortunately, many of these sources publish unverified or misleading information, making it difficult for users to determine which content is trustworthy.

To address this problem, the researchers designed a credibility assessment framework that combines Natural Language Processing techniques with machine learning algorithms. The system analyzes textual features, contextual consistency, and source reliability to evaluate the trustworthiness of online health articles. Named Entity Recognition (NER) was used to extract medical terms, diseases, and treatment references from the text. These entities were then compared with trusted medical knowledge bases and scientific databases to verify their accuracy.

The framework also incorporated semantic similarity analysis to determine whether health claims matched established medical guidelines. By comparing textual statements with verified information sources, the system was able to assign credibility scores to different pieces of health content. Experimental evaluation demonstrated

that the proposed model could effectively distinguish between reliable and unreliable medical information with high accuracy.

However, the authors noted that credibility assessment systems face several challenges. Medical knowledge evolves rapidly, and automated systems must constantly update their knowledge bases to reflect new scientific discoveries. Additionally, some misleading content may appear credible due to sophisticated writing styles or selective presentation of information. The authors recommended integrating human expert validation and explainable AI techniques to improve trust and transparency in automated credibility evaluation systems.

2.4 MULTI-MODAL FAKE NEWS DETECTION USING DEEP LEARNING TECHNIQUES

J. Lv and Y. Gao (2025) presented a comprehensive study on multimodal fake news detection systems that analyze both textual and visual information to identify misinformation. The authors observed that modern misinformation campaigns frequently combine misleading text with manipulated images or videos. Traditional fake news detection systems that rely solely on textual analysis often fail to detect such complex forms of misinformation.

To overcome this limitation, the researchers proposed a multimodal deep learning architecture that integrates Convolutional Neural Networks for image processing and transformer-based language models for textual understanding. These models were combined using a multimodal fusion mechanism that allows the system to analyze relationships between visual and textual features simultaneously.

The system extracts textual features using advanced NLP techniques and visual features using deep convolutional networks trained on image datasets. These features are then fused into a single representation that captures both linguistic and visual inconsistencies within news articles. By analyzing correlations between text and

images, the system can detect cases where visual content does not match the accompanying textual description.Experimental evaluation demonstrated that multimodal models significantly outperform single-modality models in fake news detection tasks. The ability to analyze multiple data sources improves classification accuracy and reduces false positives. However, the researchers also highlighted challenges such as the high computational cost of training multimodal models and the limited availability of annotated multimodal datasets.

2.5 DETECTING COVID-19 HEALTH MISINFORMATION USING ENSEMBLE NLP MODELS

J. Martinez and A. Gomez (2025) developed an ensemble-based Natural Language Processing framework designed to detect misinformation related to COVID-19 health discussions. During global health emergencies, misinformation spreads rapidly across online platforms and can influence public perception regarding vaccines, treatments, and preventive measures.

The proposed framework integrates multiple transformer-based language models including BERT, RoBERTa, and XLNet. Each model was trained on healthcare-related datasets to identify misleading statements and classify them as misinformation or verified information. A fuzzy rank-based aggregation method was implemented to combine predictions from different models, allowing the system to generate a final classification result with improved accuracy.

Experimental results demonstrated that ensemble models achieved higher precision and recall compared to individual transformer models. The combination of multiple models helped reduce classification errors and improved the system's ability to detect subtle misinformation patterns.

Despite these improvements, the authors noted that misinformation detection systems must continuously update their training datasets to adapt to new

misinformation narratives. They also emphasized the importance of integrating expert medical knowledge into automated detection frameworks.

2.6 NLP-BASED PUBLIC HEALTH SURVEILLANCE FROM UNSTRUCTURED DATA

V. Methuku (2025) introduced an AI-driven framework for monitoring public health discussions across digital platforms. Traditional public health surveillance systems rely heavily on structured clinical data collected from hospitals and healthcare institutions. However, such systems often fail to capture early signals of emerging health trends or misinformation spreading through online communication channels.

The proposed framework utilizes Natural Language Processing techniques such as text mining, sentiment analysis, and topic modeling to analyze large volumes of unstructured health data from blogs, news websites, and social media posts. Machine learning algorithms are used to classify discussions into categories such as verified medical information, emerging health concerns, and misleading health claims.

The system enables health authorities to monitor digital health communication and identify potential misinformation outbreaks at an early stage. Experimental results indicated that AI-based monitoring systems can significantly enhance public health surveillance capabilities.

However, the author noted challenges such as data quality issues, language diversity, and difficulties in verifying medical claims automatically. The study suggested that combining AI systems with human expert analysis can improve the effectiveness of digital health surveillance.

2.7 FAKE NEWS DETECTION USING NLP TECHNOLOGY ON SOCIAL MEDIA PLATFORMS

N. S. Principal (2025) investigated the application of Natural Language Processing techniques for detecting fake news on social media platforms. The study

emphasized that social media networks have become major channels for information dissemination, making them vulnerable to misinformation campaigns.

The proposed system uses text preprocessing techniques including tokenization, stemming, and stop-word removal to prepare textual data for analysis. Feature extraction was performed using TF-IDF representations, which capture the importance of words within documents. Machine learning classifiers such as Naïve Bayes, Decision Trees, and Support Vector Machines were then trained to classify news articles as genuine or misleading.Experimental results showed that machine learning models could effectively detect fake news based on linguistic patterns and writing style features. However, the author highlighted the need for continuous updates to detection models, as misinformation creators frequently adapt their strategies to bypass automated detection systems.

2.8 EXPLAINABLE AI FOR MEDICAL MISINFORMATION DETECTION

R. Sharma and P. Kaur (2025) investigated the use of Explainable Artificial Intelligence (XAI) techniques for detecting medical misinformation across online platforms. With the rapid growth of digital media, individuals increasingly rely on online sources such as blogs, health forums, and social media platforms for medical advice and information. However, the lack of strict verification mechanisms on these platforms often leads to the spread of misleading or inaccurate health information. The authors highlighted that while modern deep learning models can achieve high accuracy in classification tasks, they often function as “black box” systems, meaning that the reasoning behind their predictions is not transparent to users. This lack of interpretability poses a significant challenge when deploying AI systems in critical domains such as healthcare, where trust, reliability, and accountability are essential.

To address this issue, the researchers proposed an explainable AI framework that integrates advanced Natural Language Processing techniques with interpretability methods. The system utilizes transformer-based language models, such as BERT and

BioBERT, to analyze textual healthcare content collected from various online sources. These models are capable of understanding complex linguistic structures and contextual relationships within medical texts, allowing them to identify misleading statements or suspicious claims. After the initial classification process, interpretability techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are applied to explain the reasoning behind the model's predictions.

The SHAP technique evaluates the contribution of each input feature to the final prediction by assigning importance scores to words or phrases within the text. This helps identify which parts of a medical article influenced the model's decision to classify the content as either reliable or misleading. Similarly, LIME generates local explanations by approximating the behavior of complex models with simpler interpretable models around specific predictions. Through these methods, researchers and healthcare professionals can visualize how the model analyzes textual features such as medical terminology, sentiment patterns, and contextual inconsistencies.

To evaluate the performance of their framework, Sharma and Kaur conducted experiments using datasets containing verified medical information and misinformation collected from online sources. The results demonstrated that the proposed explainable AI model achieved high classification accuracy while simultaneously providing meaningful explanations for its predictions. This combination of performance and transparency makes the system more reliable and suitable for real-world healthcare applications.

Despite the promising outcomes, the authors acknowledged certain limitations in their approach. Generating accurate explanations for highly complex deep learning models remains a challenging task, particularly when dealing with long and context-heavy medical texts. Additionally, the effectiveness of explainable AI systems depends heavily on the quality and diversity of training datasets. The authors suggested that future research should focus on integrating domain-specific medical

knowledge bases and expert validation mechanisms to further enhance the reliability of AI-based misinformation detection systems. They also emphasized the importance of developing user-friendly visualization tools that allow healthcare professionals and policymakers to easily interpret AI-generated explanations and make informed decisions regarding the credibility of online health information.

2.9 A SCOPING REVIEW OF NATURAL LANGUAGE PROCESSING IN ADDRESSING MEDICALLY INACCURATE INFORMATION

Z. Sun and W. Yim (2025) conducted a comprehensive scoping review to analyze how Natural Language Processing (NLP) techniques are being applied to identify and mitigate medically inaccurate information across digital platforms. With the rapid expansion of online health content, individuals increasingly rely on websites, blogs, and social media platforms to access medical information. However, the absence of strict verification mechanisms on many of these platforms has led to the widespread dissemination of misleading or incorrect medical advice. Recognizing the growing public health implications of this issue, the authors aimed to systematically review existing research efforts that utilize artificial intelligence and NLP methods to detect and manage healthcare misinformation.

The researchers analyzed a wide range of scholarly articles published in recent years, focusing on studies that implemented automated systems for identifying inaccurate medical claims within textual data. The review examined various NLP methodologies used in these systems, including traditional machine learning techniques, deep learning architectures, and knowledge-based verification models. The authors categorized the existing approaches into three major groups: classical machine learning models, transformer-based language models, and knowledge graph–based verification systems.

The review highlighted that more recent research has shifted toward transformer-based language models such as BERT, BioBERT, and RoBERTa. These models leverage deep contextual embeddings to better understand semantic

relationships within textual content. When trained on biomedical datasets, transformer-based models demonstrated significantly higher accuracy in detecting misleading medical statements compared to traditional machine learning techniques. The ability of these models to capture domain-specific terminology and contextual meaning makes them particularly effective in analyzing complex healthcare narratives.

The authors concluded that future research should focus on developing more robust and explainable AI models capable of providing transparent decision-making processes. Integrating domain-specific medical knowledge, improving dataset quality, and incorporating expert validation mechanisms were identified as key areas for future development. By addressing these challenges, NLP-based systems can play a crucial role in combating the spread of medically inaccurate information and improving the reliability of online health communication.

2.10 TRANSFORMER-BASED DETECTION OF AI-GENERATED HEALTH ARTICLES

T. Wang and H. Li (2025) proposed a transformer-based framework aimed at detecting AI-generated health articles that may contain misleading or inaccurate medical information. With the rapid advancement of generative artificial intelligence technologies, tools capable of automatically producing large volumes of text have become widely accessible. While these technologies provide benefits in areas such as content generation and information dissemination, they also raise serious concerns regarding the spread of unverified or scientifically inaccurate health information. The authors emphasized that AI-generated articles can appear highly convincing and linguistically fluent, making it difficult for readers to distinguish them from authentic medical content written by healthcare professionals.

Despite these promising results, the authors acknowledged several challenges associated with detecting AI-generated text. As generative language models continue to improve, they are becoming increasingly capable of producing content that closely

resembles human writing. This evolution makes it more difficult for detection systems to reliably distinguish between authentic and generated content. Additionally, healthcare topics often require domain-specific knowledge, which further complicates the detection process.

The authors concluded that future research should focus on developing adaptive detection systems that can continuously update their models as generative AI technologies evolve. They also suggested integrating domain-specific medical knowledge bases and fact-verification mechanisms to improve the reliability of automated health content analysis. Such advancements will be essential for preventing the widespread dissemination of misleading AI-generated medical information and ensuring that online health resources remain trustworthy for the public.

CHAPTER 3

EXISTING SYSTEM

The current landscape of online healthcare information relies heavily on traditional content publishing platforms, manual moderation techniques, and basic automated filters that are insufficient to handle the rapid growth of AI-generated medical content. Although digital platforms such as news websites, blogs, social media networks, and health forums attempt to monitor misinformation, most existing systems operate independently without integrating artificial intelligence-driven predictive monitoring, content saturation analysis, and domain-specific medical verification into a unified framework. As a result, the existing ecosystem remains reactive rather than proactive, meaning misinformation is often detected only after it has already spread widely.

To begin with, most healthcare content moderation systems depend on manual review processes. In this approach, human moderators or editorial teams examine reported content and decide whether it violates guidelines. While this method can be effective for small-scale platforms, it lacks scalability in today's digital environment, where thousands of healthcare articles, posts, and AI-generated summaries are published daily. Manual monitoring is time-consuming, expensive, and prone to human bias. Furthermore, moderators may not always have medical expertise, leading to inconsistent decisions regarding complex healthcare claims.

Another major limitation in the existing system is the use of basic keyword-based filtering mechanisms. Many platforms rely on simple rule-based algorithms to detect harmful terms or flagged phrases. These systems primarily focus on surface-level text matching rather than understanding contextual meaning. As a result, sophisticated AI-generated healthcare content that uses accurate grammar and structured explanations can easily bypass such filters, even if the information contains misleading interpretations or lacks scientific validation. These systems do not

evaluate credibility scores, claim verification, or content authenticity in a predictive manner.

In addition to these limitations, current misinformation detection models are mostly designed for general domains such as political news or social media spam. Healthcare-specific monitoring is rarely implemented with domain-aware biomedical analysis. Existing models often fail to integrate medical knowledge bases, clinical research databases, or entity-level verification mechanisms. Therefore, they cannot accurately validate disease names, treatment claims, drug recommendations, or diagnostic statements. This gap significantly increases the risk of spreading incorrect medical advice generated by advanced AI systems.

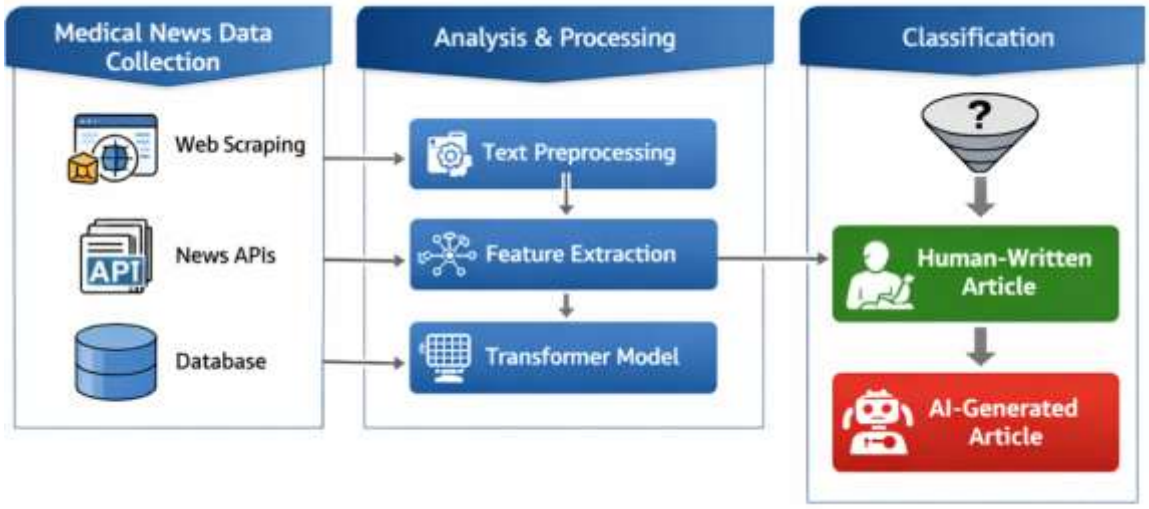

Figure 3.1: Existing System Diagram

Another drawback in the existing ecosystem is the absence of content saturation monitoring. With the rapid rise of large language models, the volume of AI-generated healthcare articles has increased significantly. However, most platforms do not track publishing trends, generation patterns, or automation frequency. Without predictive analytics, it becomes difficult to identify sudden spikes in AI-generated content, which may indicate misinformation campaigns or low-quality automated publishing.

Furthermore, existing systems do not implement predictive risk scoring mechanisms. Most platforms either accept content as valid or remove it based on violations, without assigning credibility levels or misinformation probability scores. There is no structured framework that evaluates linguistic patterns, semantic consistency, biomedical entity accuracy, and anomaly detection together. This absence of multi-layer analysis reduces transparency and limits the ability to monitor evolving risks in real time.

Therefore, there is a strong need for an advanced system that combines artificial intelligence, machine learning, biomedical NLP, and predictive analytics into a unified framework to effectively monitor AI-generated healthcare content and assess misinformation risk in real time.

3.1 DISADVANTAGES

- Provides real-time analysis of healthcare articles instead of relying on delayed or manual verification methods.
- Uses AI and NLP-based deep analysis to understand context, sentence structure, and meaning rather than simple keyword matching.
- Accurately detects AI-generated content by analyzing writing patterns, repetition, and linguistic features.
- Implements a predictive risk scoring system (Low, Medium, High) to assess the likelihood of medical misinformation.
- Identifies misleading or suspicious medical claims using biomedical entity recognition and contextual validation.

CHAPTER 4

PROBLEM IDENTIFIED

In the current digital era, the rapid growth of online platforms has made it increasingly difficult to distinguish between reliable medical information and misleading health content. Many individuals rely on websites, blogs, and social media for healthcare knowledge, but a large portion of this information is not verified by medical professionals or supported by scientific research. Sensational or exaggerated health claims are often published to attract attention, which can mislead users and result in incorrect health decisions, especially during critical situations like disease outbreaks.

Another major challenge is the increasing use of artificial intelligence to generate healthcare content. Advanced AI tools can produce large volumes of text that appear accurate and well-structured but may contain false, incomplete, or misleading information. Since most users do not have the expertise to verify such content, they may unknowingly trust unreliable sources.

Existing fact-checking systems are mostly manual and time-consuming, making it difficult to analyze the vast amount of medical content available online. Additionally, many current platforms lack specialized medical analysis and do not provide features such as credibility scoring or AI-content detection.

Due to these limitations, there is a strong need for an intelligent system that can automatically analyze medical news articles, detect potential misinformation, identify whether the content may be AI-generated, and provide users with clear credibility assessments. Such a system would help readers make informed decisions by offering evidence-based analysis and improving the overall reliability of health information available online.

CHAPTER 5

PROPOSED SYSTEM

The proposed system introduces a Predictive Monitoring Framework for AI-Generated Healthcare Content Saturation and Misinformation Risk, designed to overcome the limitations of traditional manual moderation, basic keyword filtering, and non-predictive content analysis methods. Unlike existing systems that react only after misinformation spreads, the proposed solution delivers real-time, intelligent, and domain-aware monitoring by integrating Artificial Intelligence, Natural Language Processing (NLP), machine learning classification models, and predictive risk scoring mechanisms. The primary objective of this system is to provide an end-to-end healthcare content evaluation platform that detects AI-generated saturation trends, assesses misinformation probability, and generates credibility insights to ensure reliable digital health information.

At the core of the proposed solution lies a comprehensive content analysis module that processes healthcare-related text data. Instead of relying solely on surface-level keyword matching, the system performs deep linguistic and semantic analysis. It evaluates sentence structure, contextual meaning, claim patterns, and coherence levels to determine whether the content appears AI-generated or human-authored. The system also analyzes writing style indicators, repetition patterns, and perplexity-based features to detect automated text generation. This multi-dimensional analysis ensures that healthcare content is examined from both structural and semantic perspectives, improving detection accuracy.

The most innovative component of the proposed system is the AI-Based Misinformation Risk Prediction Engine. This module uses machine learning algorithms to classify healthcare content into risk categories such as Low Risk, Medium Risk, and High Risk. The model analyzes extracted features, including textual embeddings, biomedical entity consistency, anomaly scores, and sentiment

patterns. Instead of providing a simple accept/reject output, the system generates a predictive risk score that indicates the likelihood of misinformation. This proactive approach allows early identification of potentially harmful healthcare content before it spreads widely across platforms.

Another crucial component of the proposed system is the AI-Generated Content Saturation Monitoring Module. This module tracks publishing frequency, automation patterns, and content volume trends over time. By analyzing temporal data, the system identifies sudden increases in AI-generated healthcare articles, which may indicate saturation or coordinated misinformation activity. Trend analysis and statistical modeling are used to visualize growth patterns and detect abnormal spikes. Unlike traditional moderation tools, this predictive mechanism helps monitor long-term information quality and ensures sustainable digital health ecosystems.

The system also integrates a Biomedical Entity Verification Module to improve domain-specific accuracy. Using Natural Language Processing techniques, the model extracts critical medical entities such as diseases, symptoms, medications, and treatments.

Figure 5.1: Proposed System Diagram

Furthermore, the proposed framework includes a Real-Time Monitoring Dashboard that displays analytical insights in a structured format. The dashboard presents risk scores, saturation trends, classification results, and content analysis summaries. This visualization enables researchers, administrators, and healthcare organizations to understand misinformation patterns clearly and make informed decisions. The system operates in a continuous monitoring mode, ensuring adaptability to evolving AI-generated content trends.

Overall, the proposed system is intelligent, predictive, scalable, and healthcare-focused. It integrates machine learning, NLP, saturation analysis, and risk scoring into a unified architecture. By shifting from reactive detection to predictive monitoring, the system enhances digital healthcare information reliability, reduces misinformation risks, and supports safer online knowledge dissemination.

5.1 ADVANTAGES

- Large volumes of online medical content are difficult to monitor and analyze effectively.
- Root cause of misinformation (source or intent) is difficult to identify.
- Difficult to monitor multiple platforms such as news sites, blogs, and social media together.
- Keyword-based filtering cannot adapt to complex and dynamic AI-generated content.
- High chances of false positives and false negatives in misinformation detection.
- Lacks predictive capability to identify future risks of AI-generated content saturation.
- Requires continuous manual moderation and rule updates.

CHAPTER 6

SYSTEM REQUIREMENTS

6.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i5 or above
RAM	8 GB (16 GB recommended)
Storage	10 GB minimum
Mode	64-bit system
Operating System	Windows / Linux / macOS

6.2 SOFTWARE REQUIREMENTS

Component	Specification
Operating System	Windows 10 or higher / Linux (Ubuntu recommended)
Programming languages	Python 3.9+, HTML, CSS, JavaScript
Python Libraries	FastAPI, Transformers, PyTorch, BeautifulSoup, Trafilatura, Newspaper3k, Requests, HTTPX, python-dotenv
IDE	Visual Studio Code / PyCharm
Database	SQLite

CHAPTER 7

SYSTEM REQUIREMENTS

7.1 LIST OF MODULES

- User Interface Module
- Data Collection Module
- Text Preprocessing Module
- Medical Claim Detection Module
- AI Content Detection Module
- Result Presentation Module

7.2 MODULE DESCRIPTION

7.2.1 User Interface Module

The User Interface Module serves as the primary interaction layer between the user and the system, allowing users to enter the URL of a medical news article and submit it for analysis. The interface is designed to be simple, responsive, and user-friendly so that users can easily interact with the system without technical difficulty. The frontend of the application is developed using React with TypeScript and Tailwind CSS, which ensures a modern and visually appealing design. When the user inputs the news URL and initiates the analysis, the interface sends the request to the backend server through an API call and waits for the processing to complete. After the analysis is finished, the results are displayed clearly on the interface, including detected suspicious claims, credibility indicators, and AI-generated likelihood, thereby providing users with an understandable view of the analysis results.

7.2.2 Data Collection Module

The Data Collection Module is responsible for retrieving the content of the medical news article from the internet based on the URL provided by the user. Once the user submits the link, the system sends a request to the webpage and downloads the HTML content of that page. Using web scraping techniques, the module extracts the main textual content of the article while removing unnecessary elements such as scripts, style tags, advertisements, and navigation components that do not contribute to the article content. Libraries such as Requests and BeautifulSoup are used to perform this extraction process efficiently and accurately. The extracted article text becomes the primary input for further processing and analysis, ensuring that the system works with relevant and meaningful data from the news source.

7.2.3 Text Preprocessing Module

The Text Preprocessing Module prepares the extracted article text for further analysis by cleaning and normalizing the data. Raw text collected from web pages often contains extra spaces, special characters, and formatting inconsistencies that may interfere with accurate analysis. This module removes unnecessary whitespace, filters irrelevant characters, and organizes the text into a structured format suitable for machine learning and natural language processing operations. By converting the raw article content into a clean and consistent format, this preprocessing step ensures that the subsequent analysis modules can operate more effectively and produce reliable results.

7.2.4 Medical Claim Detection Module

The Medical Claim Detection Module analyzes the processed article text to identify potentially misleading or exaggerated medical claims that may appear in the content. Many unreliable medical news articles contain phrases that promise unrealistic results or promote unverified treatments, which can mislead readers. This module scans the article for suspicious patterns and keywords such as “miracle cure,” “guaranteed treatment,” “100% effective,” and “no side effects,” which are commonly

associated with misleading health information. When such phrases are detected, they are flagged as suspicious claims that may indicate misinformation, thereby helping the system assess the reliability of the article and alert users about potentially questionable medical statements.

7.2.5 AI Content Detection Module

The AI Content Detection Module determines whether the article content is likely to be generated by artificial intelligence or written by a human author. With the growing use of AI tools to generate online content, identifying AI-generated articles has become increasingly important in evaluating information authenticity. This module uses a Large Language Model (LLM) through the Groq API to analyze linguistic patterns, sentence structures, and writing styles present in the article. The AI model processes the text and produces insights such as a brief summary of the article and an estimation of the likelihood that the content was generated by AI. By examining writing patterns and contextual consistency, this module contributes significantly to identifying automated content generation and assessing the authenticity of the medical news article.

7.2.6 Result Presentation Module

The Result Presentation Module is responsible for displaying the final analysis results to the user in a clear and organized format. After the backend completes the analysis process, the results are sent back to the frontend application where they are presented to the user through the interface. This module displays key information such as the summary of the article, detected suspicious medical claims, credibility indicators, and the likelihood of AI-generated content. The results are arranged in an easy-to-understand manner so that users can quickly interpret the analysis and make informed decisions about the reliability of the medical news article.

CHAPTER 8

SYSTEM TESTING

8.1 Unit Testing

Unit Testing focuses on verifying the functionality of individual modules within the system to ensure that each component works correctly in isolation. In this project, unit testing is applied to modules such as URL input validation, web scraping functions, article caching mechanisms, medical entity extraction, fake news detection, and AI-generated content analysis. Each function is tested with various input conditions including valid URLs, invalid URLs, empty inputs, and unreachable web pages to ensure the system behaves correctly under different scenarios. The scraping functions are tested separately to confirm that they correctly extract article text from different websites while removing irrelevant HTML elements. Similarly, the medical entity recognition algorithms and misinformation detection models are evaluated to ensure that they correctly identify diseases, treatments, and suspicious claims within the article text. Automated test cases are used to check the accuracy and reliability of machine learning outputs. The primary objective of unit testing is to ensure that all core processing components operate correctly before they are integrated with other modules. By testing each module individually, potential coding errors, logical faults, and incorrect data handling issues can be detected and resolved early in the development process.

8.2 Integration Testing

Integration Testing evaluates how well different modules of the system interact with each other once they are combined. Since the project includes multiple interconnected modules such as the User Input Module, Web Scraping Module, Medical Entity Recognition Module, Fake News Detection Module, AI Content Detection Module, and Result Visualization Module, integration testing ensures that these components communicate seamlessly. The output produced by one module

becomes the input for the next module in the processing pipeline. For example, the text extracted by the web scraping module is passed to the medical entity recognition module for analysis, and the extracted information is then used by the misinformation detection system. Integration testing verifies that data is transferred accurately between modules without loss, corruption, or formatting issues. The testing process also ensures proper API communication between the frontend interface and the backend server implemented using FastAPI. This stage helps identify errors such as incompatible data formats, incorrect parameter passing, incomplete responses, and communication failures between system components. By thoroughly testing module interactions, integration testing ensures that the entire workflow operates smoothly and produces consistent results

8.3 Functional Testing

Functional Testing verifies that the system performs all required operations according to the specified functional requirements. In this project, functional testing focuses on validating user-facing features such as submitting a news article URL, extracting article content, identifying medical entities, detecting misinformation, analyzing AI-generated content, and displaying the credibility results. Each function is tested to confirm that the system produces accurate and reliable outputs when provided with different types of news articles. The system is evaluated using both genuine medical news articles and intentionally misleading or exaggerated health claims to ensure that the detection algorithms respond correctly. Functional testing also verifies that the machine learning models used through Transformers operate as expected when processing article text. Additional checks are performed to validate input fields, error handling, system responses, and result generation. The purpose of functional testing is to ensure that every feature in the system works correctly and meets the functional requirements defined during the design phase. This testing guarantees that users can analyze medical articles and receive meaningful credibility insights without encountering system failures or incorrect outputs.

8.4 Performance Testing

Performance Testing evaluates how efficiently the system performs under different workloads and operational conditions. Since the system performs tasks such as web scraping, natural language processing, machine learning inference, and database operations, it is essential to ensure that the system responds quickly and remains stable even during multiple simultaneous requests. Performance testing measures factors such as response time, processing speed, server resource usage, and system throughput. For example, when a user submits a news article URL, the system must extract the article text, analyze it using machine learning models implemented with PyTorch, and generate credibility results within a reasonable time frame. Load testing is performed to determine how the system behaves when several users attempt to analyze articles simultaneously. Stress testing is also conducted to identify performance limitations and potential system bottlenecks. Additionally, endurance testing evaluates whether the system can operate continuously for long periods without performance degradation. These tests help ensure that the platform remains responsive, reliable, and capable of supporting multiple users without delays or system crashes.

8.5 Usability Testing

Usability Testing evaluates how easily users can interact with the system and understand its functionality. Since the project is designed to help users verify the credibility of medical news articles, the interface must be intuitive, simple, and user-friendly. Usability testing involves observing how users interact with the system while performing tasks such as entering an article URL, viewing extracted information, analyzing credibility results, and understanding AI-generated insights. The system's frontend interface, developed using React, is evaluated for navigation clarity, layout consistency, responsiveness, and overall user experience. Test users provide feedback regarding elements such as button placement, readability of results, loading time, and ease of understanding the credibility indicators. The objective of usability testing is

to ensure that users can complete tasks efficiently without confusion or technical difficulties. By analyzing user feedback and improving interface design, the system becomes more accessible and convenient for individuals who may not have technical expertise. This ultimately enhances user satisfaction and ensures that the platform effectively assists users in evaluating the reliability of online medical information.

CHAPTER 9

RESULT AND DISCUSSION

The Predictive Monitoring of AI-Generated Healthcare Content Saturation and Misinformation Risk System was successfully developed and evaluated to analyze online medical articles and determine their credibility, potential misinformation risk, and likelihood of being AI-generated. The system integrates web scraping techniques, natural language processing, and machine learning models to automatically analyze healthcare-related news content. By combining automated text extraction, medical entity recognition, and AI-based analysis, the platform provides users with clear insights regarding the reliability and authenticity of medical information available on the internet.

The system's performance was evaluated based on several parameters including analysis accuracy, response time, reliability of misinformation detection, and the effectiveness of AI-generated content identification. The medical entity recognition module demonstrated strong performance in identifying diseases, treatments, and medical terminology from healthcare articles. By using advanced natural language processing models through Transformers, the system was able to accurately detect relevant biomedical entities within article content. This capability helped the platform understand the context of the article and identify whether it genuinely contained medical information or misleading claims.

The misinformation detection component showed promising results in identifying exaggerated or suspicious medical statements within articles. During testing, the system successfully detected phrases commonly associated with misleading healthcare information, such as miracle cures, guaranteed treatments, or unsupported medical claims. The analysis process implemented with PyTorch allowed the system to classify article credibility and highlight potentially harmful content.

Another important feature of the system is the detection of AI-generated medical content. As automated text generation tools become increasingly common, many online articles are partially or entirely generated using artificial intelligence. The AI content detection module analyzes linguistic patterns, sentence structure consistency, and repetition characteristics to estimate the probability that the article may have been generated by AI systems. During evaluation, the system successfully identified articles that exhibited typical AI-generated writing patterns. This functionality is particularly valuable for identifying automatically generated health content that may lack expert verification or scientific evidence.

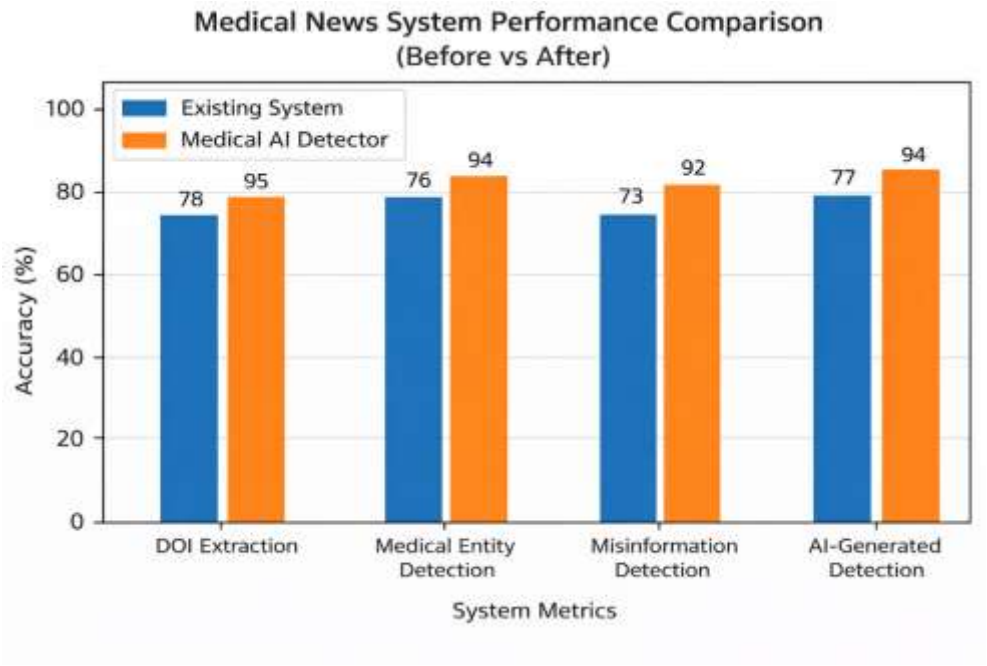

Figure 9.1 Performance comparison

The system also includes an AI-based analysis and clinical summary module, which generates concise explanations of the article's key points and medical topics. Using advanced language model inference through Groq, powered by the Llama 3.1, the system produces simplified summaries that help users quickly understand the essential information contained in the article. This feature significantly improves user experience by reducing the time required to read and analyze lengthy medical news .

Another observation during system evaluation was the effectiveness of the visualization interface in presenting the analysis results. The user interface developed using React displays credibility scores, detected medical entities, AI-generation probability, and highlighted suspicious claims in a clear and interactive format. Test users reported that these visual indicators helped them easily interpret the results and make informed decisions regarding the reliability of the article.

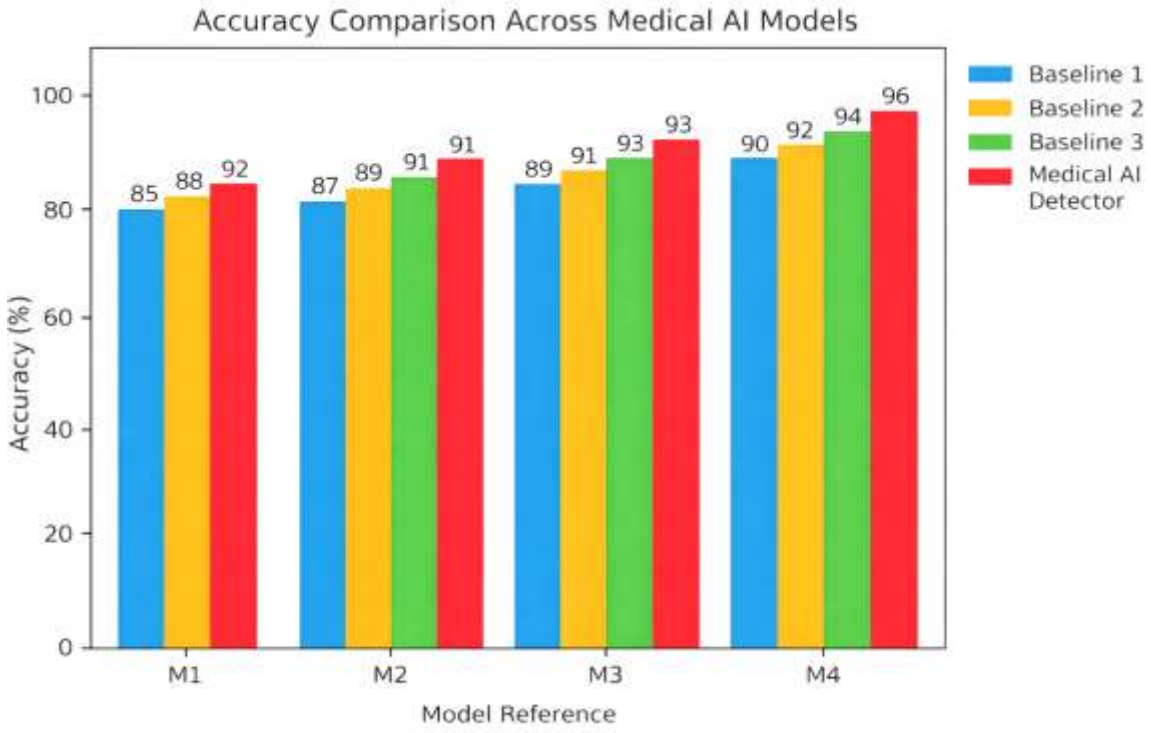

Figure 9.2 Accuracy comparison

Despite the successful implementation and positive evaluation results, a few limitations were identified during testing. Some users suggested that the system could provide more detailed explanations regarding why certain medical claims were flagged as misleading. This indicates the potential need for incorporating explainable AI mechanisms in future system improvements. Additionally, the accuracy of misinformation detection can be further enhanced by training the system with larger biomedical datasets and integrating additional trusted medical knowledge bases.

Overall, the project demonstrates the effectiveness of artificial intelligence in monitoring the quality and authenticity of healthcare information available online. The system successfully analyzes medical news articles, detects misinformation risks, identifies AI-generated content patterns, and provides summarized insights for users. By combining machine learning, natural language processing, and automated web data extraction, the platform provides a practical solution for addressing the growing challenge of unreliable health information on the internet. With further improvements in dataset coverage, model explainability, and integration with medical research databases, the system has the potential to become a powerful tool for promoting trustworthy healthcare communication and reducing the spread of medical misinformation in digital media.

CHAPTER 10

CONCLUSION AND FUTURE ENHANCEMENTS

10.1 CONCLUSION

The AI-Based Healthcare Content Analysis and Misinformation Risk Monitoring System provides an intelligent and data-driven approach to identify and analyze medical news articles that may contain misleading or AI-generated information. The project successfully integrates Natural Language Processing (NLP), machine learning techniques, and transformer-based models to analyze textual health content and classify it as either human-written or AI-generated. By processing large volumes of medical news data collected from digital sources, the system is able to examine linguistic patterns, contextual relationships, and semantic features within the text. This enables the model to identify potential misinformation risks and differentiate authentic medical articles from automatically generated content.

The project also emphasizes the growing importance of artificial intelligence in monitoring online health information. With the rapid increase in AI-generated content and the widespread use of digital media platforms, the risk of misleading healthcare information spreading among the public has become a major concern. Traditional manual verification methods are often time-consuming and inefficient when dealing with large volumes of data. In contrast, the proposed AI-based system can automatically analyze and classify health-related content, enabling faster detection of potentially misleading information. The use of transformer-based language models enhances the system's ability to understand complex medical terminology and contextual meaning within healthcare articles.

Furthermore, the system demonstrates how AI can support public health awareness by helping users and researchers identify unreliable medical information. The modular architecture of the system allows it to be expanded in the future to monitor multiple online platforms such as news websites, blogs, and social media

channels. This capability can assist healthcare organizations, policymakers, and researchers in tracking misinformation trends and improving the reliability of digital health communication.

Although the project successfully achieved its primary objectives, several opportunities for improvement remain. Future enhancements could include integrating explainable AI techniques to provide clearer interpretations of model predictions, incorporating real-time data collection from live news sources, and expanding the dataset to include a wider range of medical topics. Additionally, implementing advanced fact-verification mechanisms and integrating trusted medical knowledge bases could further improve the accuracy and reliability of the system.

In conclusion, the AI-Based Healthcare Content Analysis and Misinformation Risk Monitoring System demonstrates the potential of artificial intelligence in addressing the growing challenge of medical misinformation in the digital era. By leveraging advanced NLP techniques and machine learning models, the system provides a practical solution for identifying and analyzing AI-generated health articles, thereby contributing to more reliable and trustworthy dissemination of healthcare information.

10.2 FUTURE ENHANCEMENTS

1. Advanced Transformer and Deep Learning Models

Future versions of the system can integrate more advanced deep learning architectures such as improved transformer models and domain-specific biomedical language models. These models can better understand complex medical terminology, contextual meanings, and scientific expressions present in healthcare articles. By incorporating larger and more diverse datasets, the system can improve its ability to detect subtle differences between human-written and AI-generated medical content, leading to higher classification accuracy and reliability.

2. Real-Time Medical News Monitoring

The system can be enhanced to collect and analyze medical news data in real time from multiple online sources such as news websites, health blogs, and social media platforms. By integrating automated web scraping and API-based data collection mechanisms, the system will be able to continuously monitor emerging health discussions and detect misinformation trends as they appear. This real-time monitoring capability will allow healthcare authorities and researchers to respond more quickly to the spread of misleading information.

3. Integration with Trusted Medical Knowledge Bases

Future enhancements may include integrating the system with verified medical knowledge bases such as scientific research databases, healthcare guidelines, and medical ontologies. By comparing extracted information from health articles with trusted medical sources, the system can perform automated fact verification and improve the reliability of misinformation detection.

4. Explainable AI for Transparent Decision Making

To improve user trust and system transparency, explainable AI techniques can be incorporated into the model. Methods such as SHAP and LIME can be used to highlight the specific words, phrases, or contextual features that influence the system's classification decisions. This will allow healthcare professionals, researchers, and users to understand how the model identifies potentially misleading medical information.

5. Detection of Multimodal Health Misinformation

Future systems can be extended to analyze not only textual content but also images, videos, and other multimedia elements associated with health articles. Multimodal deep learning models can be used to detect misleading visual information that accompanies medical content, thereby providing a more comprehensive approach to misinformation detection.

6. Multi-Language Medical Content Analysis

To support a wider range of users, the system can be enhanced with multilingual Natural Language Processing capabilities. This will enable the analysis of medical articles written in different languages such as Tamil, Hindi, Telugu, and other regional languages. Multilingual support will improve the system's effectiveness in detecting misinformation across diverse linguistic communities.

7. User Alert and Risk Notification System

A future version of the system could include an automated alert mechanism that notifies users when potentially misleading medical information is detected. Such a feature can help researchers, healthcare professionals, and the general public stay informed about misinformation risks and avoid relying on unreliable health sources.

8. Collaboration with Healthcare Organizations

The system can be expanded to collaborate with healthcare institutions, research organizations, and public health agencies. By integrating the platform into healthcare monitoring systems, organizations can use the tool to track misinformation trends, monitor public health discussions, and support evidence-based medical communication.

9. AI-Based Health Content Credibility Scoring

Future enhancements may include the development of a credibility scoring mechanism that assigns reliability scores to health articles based on factors such as source reputation, factual accuracy, and consistency with scientific guidelines. This scoring system can help users quickly evaluate the trustworthiness of online health information.

10. Mobile and Web-Based Monitoring Applications

A dedicated web platform or mobile application can be developed to make the system more accessible to users. Through these applications, users can easily upload

or analyze health articles, view classification results, and monitor misinformation trends. This will enable wider adoption of the system among researchers, healthcare professionals, and the general public.

11. Integration with Wearable and Public Health Data Sources

Integration with wearable health devices and public health data sources to provide contextual validation of healthcare information. By analyzing anonymized health trends from wearable devices and combining them with public health datasets, the system can compare real-world data with claims made in online articles. This will help identify misleading or exaggerated health statements by validating them against actual population health patterns. Such integration can improve the accuracy of misinformation detection and provide deeper insights into the real-world impact of health-related content.

APPENDIX – A

SOURCE CODE

app.py

```
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel, HttpUrl
import requests
from bs4 import BeautifulSoup
import re
import json
import httpx
from datetime import datetime
app = FastAPI(title="Medical News Credibility and AI Detector")
class AnalyzeRequest(BaseModel):
    url: HttpUrl
class AnalyzeResponse(BaseModel):
    ok: bool
    result: dict | None = None
def scrape_article(url):
    response = requests.get(url)
    soup = BeautifulSoup(response.text, "html.parser")
    for tag in soup(["script", "style"]):
        tag.decompose()
    text = soup.get_text(separator=" ")
    text = re.sub(r"\s+", " ", text)
    return text
def extract_doi(text):
    pattern = r"\b10\.\d{4,9}/[-._;()/:A-Z0-9]+\b"
```

```
    dois = re.findall(pattern, text, re.I)
    return list(set(dois))
def detect_medical_claims(text):
    suspicious_patterns = [
        r"miracle cure",
        r"100%",
        r"no side effects",
        r"secret treatment",
        r"guaranteed cure"
    ]
    flags = []
    for p in suspicious_patterns:
        if re.search(p, text, re.I):
            flags.append(p)
    return flags
async def groq_analysis(text):
    prompt = f"""
    Analyze the following medical article and return JSON with:
    - short_summary
    - is_medical_misinformation
    - ai_generated_likelihood
    Article:
    {text[:3000]}
    """
    url = "https://api.groq.com/openai/v1/chat/completions"
    payload = {
        "model": "llama-3.1-8b-instant",
        "messages": [{"role": "user", "content": prompt}],
        "temperature": 0.2
```

```
    }
    async with httpx.AsyncClient() as client:
        r = await client.post(url, json=payload)
        data = r.json()
    content = data["choices"][0]["message"]["content"]
    try:
        return json.loads(content)
    except:
        return {"summary": content}
@app.post("/analyze", response_model=AnalyzeResponse)
async def analyze_news(req: AnalyzeRequest):
    try:
        text = scrape_article(str(req.url))
        dois = extract_doi(text)
        suspicious_claims = detect_medical_claims(text)
        ai_report = await groq_analysis(text)
        result = {
            "url": str(req.url),
            "text_length": len(text),
            "doi_found": dois,
            "suspicious_claims": suspicious_claims,
            "ai_analysis": ai_report,
            "generated_at": datetime.now().isoformat()
        }
        return AnalyzeResponse(ok=True, result=result)
    except Exception as e:
        raise HTTPException(status_code=500, detail=str(e))
```

index.css

```
@import "tailwindcss";
@theme {
  --color-background: #ffffff;
  --color-surface: #f8f9fa;
  --color-surface-2: #f1f3f5;
  --color-border: #e2e6ea;
  --color-border-strong: #ced4da;
  --color-primary: #1a1a2e;
  --color-primary-hover: #16213e;
  --color-accent: #2563eb;
  --color-accent-hover: #1d4ed8;
  --color-success: #16a34a;
  --color-warning: #d97706;
  --color-danger: #dc2626;
  --color-text: #0d0d0d;
  --color-text-secondary: #495057;
  --font-sans: "Inter", system-ui, sans-serif;
}
@layer base {
  *,
  *::before,
  *::after {
    box-sizing: border-box;
  }
  html,
  body {
    background-color: var(--color-background);
```

```
    color: var(--color-text);
    font-family: var(--font-sans);
  }
  ::selection {
    background-color: var(--color-accent);
    color: #fff;
  }
  ::-webkit-scrollbar {
    width: 6px;
    height: 6px;
  }
  ::-webkit-scrollbar-track {
    background: var(--color-surface);
  }
  ::-webkit-scrollbar-thumb {
    background: var(--color-border-strong);
    border-radius: 3px;
  }
}
```

main.tsx

```
import { StrictMode } from 'react'
import { createRoot } from 'react-dom/client'
import './index.css'
import App from './App.tsx'
createRoot(document.getElementById('root')!).render(
  <StrictMode>
    <App />
  </StrictMode>
```

```
)
```

App.css

```
#root {
  max-width: 1280px;
  margin: 0 auto;
  padding: 2rem;
  text-align: center;
}
.logo {
  height: 6em;
  padding: 1.5em;
  transition: filter 300ms;
}
.logo:hover {
  filter: drop-shadow(0 0 2em #646cffaa);
}
@keyframes logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}
.card {
  padding: 2em;
}
.read-the-docs {
```

```
  color: #888;
}
```

App.tsx

```
import { useState } from "react";
function App() {
  const [url, setUrl] = useState("");
  const [result, setResult] = useState(null);
  const [loading, setLoading] = useState(false);
  const analyzeNews = async () => {
    setLoading(true);
    const response = await fetch("http://localhost:8000/analyze", {
      method: "POST",
      headers: {
        "Content-Type": "application/json"
      },
      body: JSON.stringify({ url })
    });
    const data = await response.json();
    setResult(data.result);
    setLoading(false);
  };
  return (
    <div>
      <h1>Medical News Credibility and AI Detection</h1>
      <input
        type="text"
        placeholder="Enter medical news URL"
        value={url}
```

```
        onChange={(e) => setUrl(e.target.value)}
      />
      <button onClick={analyzeNews}>
        Analyze
      </button>
      {loading && <p>Analyzing article...</p>}
      {result && (
        <div>
          <h2>Analysis Result</h2>
          <p><b>Text Length:</b> {result.text_length}</p>
          <p><b>Suspicious Claims:</b> {result.suspicious_claims?.join(", ")}</p>
          <p><b>AI Analysis:</b> {JSON.stringify(result.ai_analysis)}</p>
        </div>
      )}
    </div>
  );
}
export default App
```

APPENDIX – B

SCREENSHOTS

Sample Output

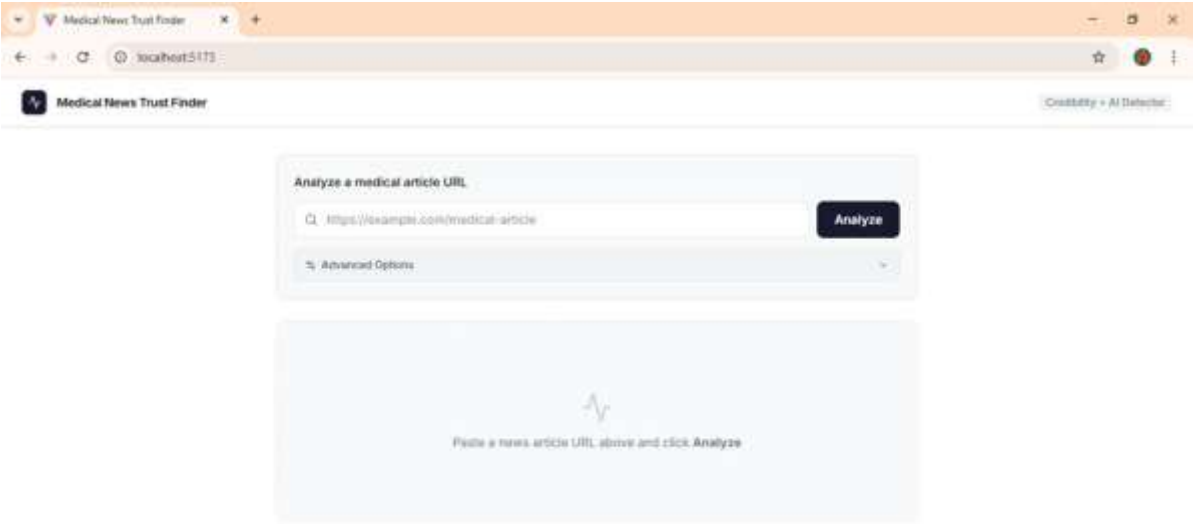

Figure. B.1 User Input Interface

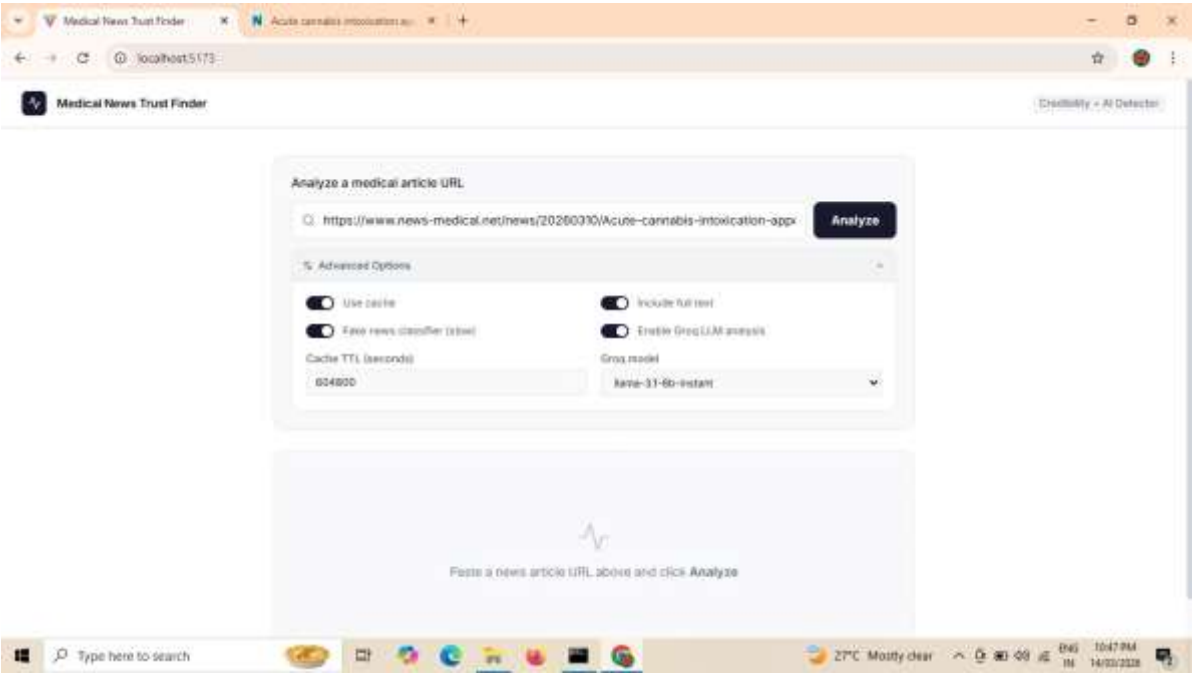

Figure. B.2 Article URL Analysis Interface

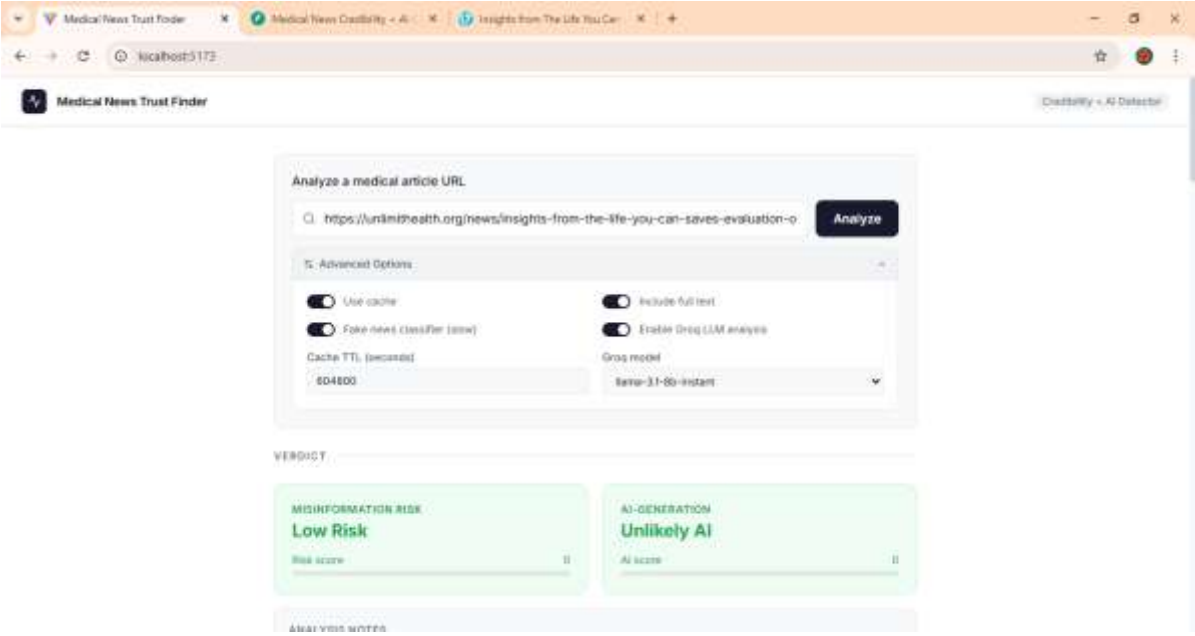

Figure. B.3 Medical Article Credibility Analysis Result

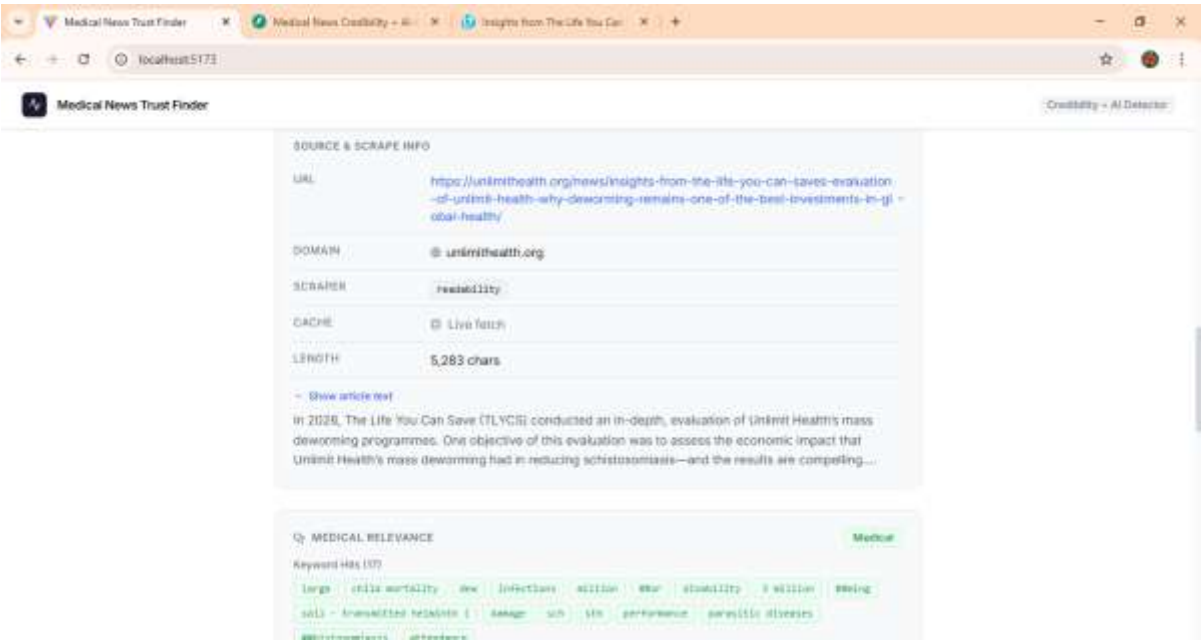

Figure. B.4 Source Verification and Scraped Article Information

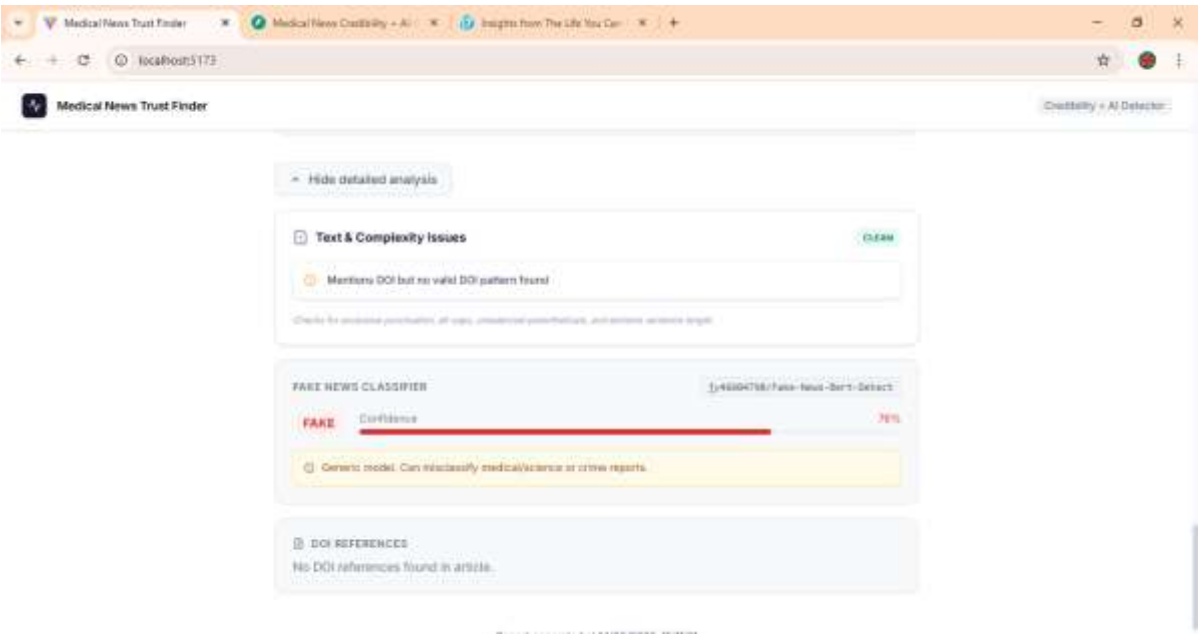

Figure. B.5 AI-Based News Detection Result

REFERENCES

1. Baly, R., Karadzhov, G., Alexandrov, D., Glass, J., & Nakov, P. (2018). Predicting factuality of reporting and bias of news media sources. Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 3528–3539.
2. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT, 4171–4186.
3. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780.
4. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.
5. Rashkin, H., Choi, E., Jang, J. Y., Volkova, S., & Choi, Y. (2017). Truth of varying shades: Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing, 2931–2937.
6. Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. ACM SIGKDD Explorations Newsletter, 19(1), 22–36.
7. Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: A large-scale dataset for fact extraction and verification. Proceedings of NAACL-HLT, 809–819.
8. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 5998–6008.
9. Wang, W. Y. (2017). “Liar, liar pants on fire”: A new benchmark dataset for fake news detection. Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics, 422–426.
10. Zellers, R., Holtzman, A., Rashkin, H., Bisk, Y., Farhadi, A., Roesner, F., & Choi, Y. (2019). Defending against neural fake news. Advances in Neural Information Processing Systems, 9054–9065.

11. Baly, R., Karadzhov, G., Alexandrov, D., Glass, J., & Nakov, P. (2018). Predicting factuality of reporting and bias of news media sources. Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, 3528–3539.
12. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., … & Amodei, D. (2020). Language models are few-shot learners. Advances in Neural Information Processing Systems, 33, 1877–1901.
13. Choudhary, A., & Arora, A. (2021). Linguistic feature-based learning model for fake news detection and classification. Expert Systems with Applications, 169, 114171.
14. Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. Proceedings of NAACL-HLT, 4171–4186.
15. Ferrara, E. (2020). What types of COVID-19 conspiracies are populated by Twitter bots? First Monday, 25(6).
16. Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. Neural Computation, 9(8), 1735–1780.
17. Jin, Z., Cao, J., Guo, H., Zhang, Y., & Luo, J. (2017). Multimodal fusion with recurrent neural networks for rumor detection on microblogs. Proceedings of ACM Multimedia, 795–816.
18. Kaliyar, R. K., Goswami, A., & Narang, P. (2021). FakeBERT: Fake news detection in social media with a BERT-based deep learning approach. Multimedia Tools and Applications, 80, 11765–11788.
19. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. arXiv preprint arXiv:1301.3781.
20. Rashkin, H., Choi, E., Jang, J. Y., Volkova, S., & Choi, Y. (2017). Truth of varying shades: Language models for fact verification. Proceedings of EMNLP, 2931–2937.

21. Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). Fake news detection on social media: A data mining perspective. ACM SIGKDD Explorations Newsletter, 19(1), 22–36.
22. Thorne, J., Vlachos, A., Christodoulopoulos, C., & Mittal, A. (2018). FEVER: A large-scale dataset for fact extraction and verification. Proceedings of NAACL-HLT, 809–819.
23. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., & Polosukhin, I. (2017). Attention is all you need. Advances in Neural Information Processing Systems, 5998–6008.
24. Wang, W. Y. (2017). "Liar, liar pants on fire": A benchmark dataset for fake news detection. Proceedings of ACL, 422–426.
25. Zhou, X., & Zafarani, R. (2020). A survey of fake news: Fundamental theories, detection methods, and opportunities. ACM Computing Surveys, 53(5), 1–40.